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Integrated circuit package and method of forming same

Abstract

An embodiment is a package including a package substrate and a package component bonded to the package substrate, the package component including an interposer, an optical die bonded to the interposer, the optical die including an optical coupler, an integrated circuit die bonded to the interposer adjacent the optical die, a lens adapter adhered to the optical die with a first optical glue, a mirror adhered to the lens adapter with a second optical glue, the mirror being aligned with the optical coupler of the optical die, and an optical fiber on the lens adapter, a first end of the optical fiber facing the mirror, the optical fiber being configured such that an optical data path extends from the first end of the optical fiber through the mirror, the second optical glue, the lens adapter, and the first optical glue to the optical coupler of the optical die.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of U.S. Provisional Application No. 63/379,015, filed on Oct. 11, 2022 and entitled “LAD (Lens Adopter) & PML (Photonic Mirror Lens) Design and Construction for CPO (Co-Package Optics) Application”, and U.S. Provisional Application No. 63/374,798, filed Sep. 7, 2022, and entitled “LAD (Lens Adopter) & PML (Photonic Mirror Lens) Design and Construction for CPO (Co-Package Optics) Application”, which application is hereby incorporated herein by reference.

BACKGROUND

(1) The semiconductor industry has experienced rapid growth due to ongoing improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, improvement in integration density has resulted from iterative reduction of minimum feature size, which allows more components to be integrated into a given

area. As the demand for shrinking electronic devices has grown, a need for smaller and more creative packaging techniques of semiconductor dies has emerged.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional view of an integrated circuit die, in accordance with some embodiments.

(3) FIG. 2 illustrates a cross-sectional view of an integrated circuit die, in accordance with some embodiments.

(4) FIG. 3 illustrates a cross-sectional view of an integrated circuit die, in accordance with some embodiments.

(5) FIGS. 4-9, 10A, 10B, and 11-15 illustrate plan and cross-sectional views of intermediate stages in the manufacturing of package components, in accordance with some embodiments.

(6) FIGS. 16A, 16B, 17A, 17B, 17C, 17D, 17E, 17F, 17G, 17H, 18A, 18B, and 18C illustrate magnified cross-sectional views of a package component, in accordance with some embodiments.

(7) FIG. 19 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

(8) FIG. 20 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

(9) FIG. 21 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

(10) FIG. 22 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

(11) FIG. 23 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

(12) FIG. 24 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

(13) FIG. 25 illustrates a cross-sectional view of a package component, in accordance with some embodiments.

DETAILED DESCRIPTION

(14) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(15) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are

intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(16) Embodiments will be described with respect to embodiments in a specific context, namely integrated circuit packages and methods of forming the same. In various embodiments presented herein, a package comprises a package component mounted on a package substrate. The package component may comprise an optical integrated circuit die attached to a redistribution structure or an interposer. The optical integrated circuit die may comprise an optical coupler, such as a grating coupler. Various embodiments presented herein allow for integration of optical integrated circuit dies comprising grating couplers, achieving high bandwidth with ultra-low power consumption through the grating coupler, and extensive integration for co-package. The embodiments include a lens adapter, a support structure, and a mirror. By including these components in specific configurations, the package structure can be more compact and utilize a lateral entry optical fiber to the package structure with a top surface entry to the optical engine for the optical data path from the optical fiber. Further, the optical loss from the optical fiber to the mirror is reduced as compared to other structures. In addition, the lens adapter and the mirror allow for the disclosed embodiments to be widely usable in various packaging configurations, such as multi-chip modules (MCM), chip-on-wafer-on-substrate packages, or integrated fan-out (InFO) packages.

(17) FIG. 1 illustrates a cross-sectional view of an integrated circuit die **50**, in accordance with some embodiments. Integrated circuit dies **50** will be packaged in subsequent processing to form integrated circuit packages. Each integrated circuit die **50** may be a logic device (e.g., application-specific integrated circuit (ASIC) die, central processing unit (CPU), graphics processing unit (GPU), microcontroller, etc.), a memory device (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), a power management device (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) device, a sensor device, a micro-electro-mechanical-system (MEMS) device, a signal processing device (e.g., digital signal processing (DSP) die), a front-end device (e.g., analog front-end (AFE) dies), the like, or a combination thereof (e.g., a system-on-a-chip (SoC) die). The integrated circuit die **50** may be formed in a wafer, which may include different die regions that are singulated in subsequent steps to form a plurality of integrated circuit dies **50**. The integrated circuit die **50** includes a semiconductor substrate **52**, an interconnect structure **54**, and conductive connectors **56**.

(18) The semiconductor substrate **52** may be a substrate of silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate **52** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The semiconductor substrate **52** has an active or a front-side surface (e.g., the surface facing upward) and an inactive or a backside surface (e.g., the surface facing downward). Devices are at the active surface of the semiconductor substrate **52**. The devices may be active devices (e.g., transistors, diodes, etc.) and/or passive devices (capacitors, resistors, inductors, etc.). The inactive surface may be free from devices.

(19) The interconnect structure **54** is over the active surface of the semiconductor substrate **52**, and is used to electrically connect the devices of the semiconductor substrate **52** to form an integrated circuit. The interconnect structure **54** may include one or more dielectric layer(s) and respective metallization layer(s) in the dielectric layer(s). Acceptable dielectric materials for the dielectric layers include low-k dielectric materials such as phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like.

Acceptable dielectric materials for the dielectric layers further include oxides such as silicon oxide or aluminum oxide; nitrides such as silicon nitride; carbides such as silicon carbide; the like; or combinations thereof such as silicon oxynitride, silicon oxycarbide, silicon carbonitride, silicon oxycarbonitride or the like. Other dielectric materials may also be used, such as a polymer such as polybenzoxazole (PBO), polyimide, a benzocyclobuten (BCB) based polymer, or the like. The metallization layers may include conductive vias and/or conductive lines to interconnect the devices of the semiconductor substrate **52**. The metallization layers may be formed of a conductive material, such as a metal, such as copper, cobalt, aluminum, gold, combinations thereof, or the like. The interconnect structure **54** may be formed by a damascene process, such as a single damascene process, a dual damascene process, or the like.

(20) Conductive connectors **56** are formed at the front-side **50F** of the integrated circuit die **50**. The conductive connectors **56** may comprise underbump metallizations (UBMs) **56A** and solder regions **56B** over the UBMs **56A**. The UBMs **56A** may be conductive pillars, pads, or the like. In some embodiments, the UBMs **56A** may be formed by forming a seed layer over the interconnect structure **54**. The seed layer may be a metal layer, which may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. In some embodiments, the seed layer includes a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the UBMs **56A**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is then formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may include a metal, such as copper, titanium, tungsten, aluminum, nickel, or the like. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process. The remaining portions of the seed layer and conductive material form the UBMs **56A**.

(21) In some embodiments, the UBMs **56A** may include three layers of conductive materials, such as a layer of titanium, a layer of copper, and a layer of nickel. Other arrangements of materials and layers, such as an arrangement of chrome/chrome-copper alloy/copper/gold, an arrangement of titanium/titanium tungsten/copper, or an arrangement of copper/nickel/gold, may be utilized for the formation of the UBMs **56A**. Any suitable materials or layers of material that may be used for the UBMs **56A** are fully intended to be included within the scope of the current application.

(22) The solder regions **56B** may comprise a solder material and may be formed over the UBMs **56A** by dipping, printing, plating, or the like. The solder material may comprise, for example, lead-based and lead-free solders, such as Pb—Sn compositions for lead-based solder; lead-free solders including InSb; tin, silver, and copper (SAC) compositions; and other eutectic materials that have a common melting point and form conductive solder connections in electrical applications. For lead-free solder, SAC solders of varying compositions may be used, such as SAC 105 (Sn 98.5%, Ag 1.0%, Cu 0.5%), SAC 305, and SAC 405, as examples. Lead-free solders may further include SnCu compounds as well, without the use of silver (Ag). Lead-free solders may also include tin and silver, Sn—Ag, without the use of copper. In some embodiments, a reflow process may be performed, giving the solder regions **56B** a shape of a partial sphere in some embodiments. In other embodiments, the solder regions **56B** may have other shapes, such as non-spherical shapes.

(23) In some embodiments, the solder regions **56B** may be used to perform chip probe (CP) testing on the integrated circuit die **50**. For example, the solder regions may be solder balls, solder bumps, or the like, which are used to attach a chip probe to the conductive connectors **56**. Chip probe testing may be performed on the integrated circuit die **50** to ascertain whether the integrated circuit

die **50** is a known good die (KGD). Thus, only integrated circuit dies **50**, which are KGDs, undergo subsequent processing and are packaged, and dies which fail the chip probe testing are not packaged. In some embodiments, after testing, the solder regions **56B** may be removed in subsequent processing steps.

(24) FIG. 2 illustrates a cross-sectional view of an integrated circuit die **60**, in accordance with some embodiments. The integrated circuit die **60** is a stacked device that includes multiple semiconductor substrates **52**. For example, the integrated circuit die **60** may be a memory device that includes multiple memory dies such as a hybrid memory cube (HMC) device, a high bandwidth memory (HBM) device, or the like. In such embodiments, the integrated circuit die **60** includes multiple semiconductor substrates **52** interconnected by through-substrate vias (TSVs) such as through-silicon vias (not shown). Each of the semiconductor substrates **52** may (or may not) have a separate interconnect structure.

(25) FIG. 3 illustrates a cross-sectional view of an integrated circuit die **70**, in accordance with some embodiments. The integrated circuit die **70** may be an optical integrated circuit die, such as an optical engine die. The integrated circuit die **70** may include an electrical integrated circuit (EIC) **70A** bonded to a photonic integrated circuit (PIC) **70B**. The EIC **70A** may comprise a semiconductor substrate **52**, active and/or passive electric devices on the active side of the semiconductor substrate **52**, an interconnect structure **54** on the active side of the semiconductor substrate **52**, and a lens **74** on the backside of the semiconductor substrate **52**. The lens **74** may be formed by patterning the backside of the semiconductor substrate **52**. The lens **74** may be formed to protrude from the backside of the semiconductor substrate **52** (see, e.g., FIG. 3) or to be recessed into the backside of the semiconductor substrate **52** (see, e.g., FIG. 25). The EIC **70A** may be formed in a similar manner as the integrated circuit die **50** described above with reference to FIG. 1, and the description is not repeated herein.

(26) The PIC **70B** may comprise optical devices, such as waveguides, modulators, or the like. The PIC **70B** may also include an optical coupler **72**, such as a grating coupler. In some embodiments, the optical coupler **72** may comprise a dielectric material (such as, silicon nitride, or the like) and may be formed using physical vapor deposition (PVD), chemical vapor deposition (CVD), or the like. In other embodiments, the optical coupler **72** may comprise a semiconductor layer (such as, a silicon layer, or the like) and may be formed from an SOI substrate. The optical coupler **72** may be disposed within the PIC **70B**. As described below with greater detail, the optical coupler **72** provides optical coupling between the integrated circuit die **70** and an optical fiber coupled to the integrated circuit die **70**.

(27) The integrated circuit die **70** may be formed in a wafer, which may include different die regions that are singulated in subsequent steps to form a plurality of integrated circuit dies **70**. In some embodiments, the wafer may be formed by hybrid bonding an EIC wafer (comprising a plurality of EICs **70A**) to a PIC wafer (comprising a plurality of PICs **70B**).

(28) FIGS. 4-15 illustrate top and cross-sectional views of intermediate stages in the manufacturing of package components **600**, in accordance with some embodiments. The package components **400** may be chip-on-wafer (CoW) package components. FIGS. 4-10A and 11-15 illustrate cross-sectional views, and FIG. 10B illustrates a plan view. In particular, FIGS. 4-6 illustrate formation of a wafer-level package component **200**, in accordance with some embodiments. In some embodiments, the wafer-level package component **200** comprises a plurality of package regions (such as a region **200A**) that correspond to package components (such as package components **400**). The plurality of package regions of the wafer-level package component **200** are singulated to form individual packaged components **400** as described below in FIG. 7.

(29) In FIG. 4, an interposer wafer **200** is obtained or formed. The interposer wafer **200** comprises a plurality of package regions, such as the package region **200A**. The interposer wafer **200** comprises an interposer **202** in a package region (such as the package region **200A**), which will be singulated in subsequent processing to be included in the package component **400**. In some

embodiments, the interposers **202** include a substrate **204**, an interconnect structure **206**, and conductive vias **208**.

(30) The substrate **204** may be formed using similar materials and methods as the semiconductor substrate **52** described above with reference to FIG. **1**, and the description is not repeated herein. In some embodiments, the substrate **204** generally does not include active devices therein, although the interposers **202** may include passive devices formed in and/or on an active or a front surface (e.g., the surface facing upward in FIG. **4**) of the substrate **204**. In other embodiments, active devices such as transistors, capacitors, resistors, diodes, and the like, may be formed in and/or on the front surface of the substrate **204**.

(31) The interconnect structure **206** is formed over the front surface of the substrate **204**, and is used to electrically connect the devices (if any) of the substrate **204**. The interconnect structure **206** may include one or more dielectric layer(s) and respective metallization layer(s) in the dielectric layer(s). The interconnect structure **206** may be formed using similar materials and methods as the interconnect structure **54** described above with reference to FIG. **1**, and the description is not repeated herein.

(32) Further in FIG. **4**, conductive connectors **120** are formed on the front-side **200FS** of the interposer wafer **200** and in electrical contact with the interconnect structure **206**. The conductive connectors **120** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **120** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **120** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **120** comprise metal pillars (such as copper pillars) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

(33) In some embodiments, the conductive connectors **120** comprise UBMs **120A** and solder regions **120B** over the UBMs **120A**. The UBMs **120A** may be formed using similar materials and methods as the UBMs **56A** described above with reference to FIG. **1**, and the description is not repeated herein. The solder regions **120B** may be formed using similar materials and methods as the solder regions **56B** described above with reference to FIG. **1**, and the description is not repeated herein.

(34) The conductive vias **208** extend into the interconnect structure **206** and/or the substrate **204**. The conductive vias **208** are electrically connected to metallization layer(s) of the interconnect structure **206**. The conductive vias **208** are also sometimes referred to as through substrate vias (TSVs). As an example to form the conductive vias **208**, recesses can be formed in the interconnect structure **206** and/or the substrate **204** by, for example, etching, milling, laser techniques, a combination thereof, and/or the like. A thin dielectric material may be formed in the recesses, such as by using an oxidation technique. A thin barrier layer may be conformally deposited in the openings, such as by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), thermal oxidation, a combination thereof, and/or the like. The barrier layer may be formed of an oxide, a nitride, a carbide, combinations thereof, or the like. A conductive material may be deposited over the barrier layer and in the openings. The conductive material may be formed by an electro-chemical plating process, CVD, ALD, PVD, a combination thereof, and/or the like. Examples of conductive materials are copper, tungsten, aluminum, silver, gold, a combination

thereof, and/or the like. Excess conductive material and barrier layer is removed from a surface of the interconnect structure **206** or the substrate **204** by, for example, a chemical mechanical polishing (CMP). Remaining portions of the barrier layer and conductive material form the conductive vias **208**.

(35) In FIG. 5, integrated circuit dies **50**, **60** and **70** are attached to the interconnect structure **206**. In the illustrated cross-sectional view, each package region (such as the region **200A** illustrated in FIG. 5) of the wafer-level package component **200** comprises a single integrated circuit die **50**, a single integrated circuit die **60**, and a single integrated circuit die **70**. The integrated circuit die **50** may be a logic device, such as an application-specific integrated circuit (ASIC) die, a central processing unit (CPU), a graphics processing unit (GPU), a system-on-a-chip (SoC), a microcontroller, or the integrated circuit die **50** may be a memory device, such as a dynamic random access memory (DRAM) die, a static random access memory (SRAM) die, or the like. In some embodiments, there may be multiple integrated circuit dies **50** and one may be a logic device and the other may be a memory device. In some embodiments, multiple integrated circuit dies **50** may be included and may be the same type of dies, such as SoC dies, ASIC dies, or the like. Although a single integrated circuit die **60** and a single integrated circuit die **70** are shown in the cross-sectional view in FIG. 5, there may be a plurality of integrated circuit dies **60** and a plurality of single integrated circuit dies **70** in each package region (such as the region **200A**) of the wafer-level package component **200**.

(36) In some embodiments, the integrated circuit dies **50**, **60** and **70** are attached to the interconnect structure **206** using the conductive connectors **56** (see FIGS. 1-3) and **120**. The integrated circuit dies **50**, **60** and **70** may be placed on the interconnect structure **206** using, e.g., a pick-and-place tool. After placing the integrated circuit dies **50**, **60** and **70** on the interconnect structure **206**, the solder regions **56B** of the conductive connectors **56** (see FIGS. 1-3) are in physical contact with respective solder regions **120B** of respective conductive connectors **120**. After placing the integrated circuit dies **50**, **60** and **70** on the interconnect structure **206**, a reflow process is performed on the conductive connectors **56** and **120** (see FIGS. 1-4). The reflow process melts and merges the solder regions **56B** and **120B** into solder joints **124**. The solder joints **124** electrically and mechanically couple the integrated circuit dies **50**, **60** and **70** to the interconnect structure **206**.

(37) Further in FIG. 5, an underfill **126** may be formed around the solder joints **124**, and in a gap between the interconnect structure **206** and the integrated circuit dies **50**, **60** and **70**. The underfill **126** may reduce stress and protect the solder joints **124**. The underfill **126** may be formed of an underfill material such as a molding compound, epoxy, or the like. The underfill **126** may be formed by a capillary flow process after the integrated circuit dies **50**, **60** and **70** are attached to the interconnect structure **206**, or may be formed by a suitable deposition method before the integrated circuit dies **50**, **60** and **70** are attached to the interconnect structure **206**. The underfill **126** may be applied in liquid or semi-liquid form and then subsequently cured. In some embodiments, the underfill **126** partially or fully fills gaps between adjacent ones of the integrated circuit dies **50**, **60** and **70**, such that the underfill **126** extends along sidewalls of the integrated circuit dies **50**, **60** and **70**.

(38) Further in FIG. 5, an encapsulant **132** is formed on and around the integrated circuit dies **50**, **60**, and **70**. After formation, the encapsulant **132** encapsulates the integrated circuit dies **50**, **60** and **70**, and the underfill **126**. The encapsulant **132** may be a molding compound, epoxy, or the like. The encapsulant **132** may not include fillers therein. The encapsulant **132** may be applied by compression molding, transfer molding, or the like, and is formed over the package component **200** such that the integrated circuit dies **50**, **60** and **70** are buried or covered. The encapsulant **132** may be applied in liquid or semi-liquid form and then subsequently cured. The encapsulant **132** may be thinned to expose the integrated circuit dies **50**, **60** and **70**. The thinning process may be a grinding process, a CMP, an etch-back, combinations thereof, or the like. After the thinning process, top surfaces of the integrated circuit dies **50**, **60** and **70**, and the encapsulant **132** are coplanar (within

process variations), such that they are level with one another. The thinning is performed until a desired amount of the integrated circuit dies **50**, **60** and **70**, and/or the encapsulant **132** has been removed.

(39) In FIG. **6**, the wafer-level package component of FIG. **5** is flipped over is attached to a carrier wafer **210**. The carrier wafer **210** is used as a platform or a support for a packaging process described below. In some embodiments, the carrier wafer **210** comprises a semiconductor material (such as silicon, or the like), a dielectric material (such as glass, a ceramic material, quartz, or the like), a combination thereof, or the like. In some embodiments, the wafer-level package component is attached to the carrier wafer **210** using an adhesive (not shown).

(40) The substrate **204** may be thinned to expose the conductive vias **208**. Exposure of the conductive vias **208** may be accomplished by a thinning process, such as a grinding process, a CMP, an etch-back, combinations thereof, or the like. In some embodiments (not separately illustrated), the thinning process for exposing the conductive vias **208** includes a CMP, and the conductive vias **208** protrude at the back-side **200BS** of the wafer **200** as a result of dishing that occurs during the CMP. In such embodiments, an insulating layer (not separately illustrated) may optionally be formed on the back surface of the substrate **204**, surrounding the protruding portions of the conductive vias **208**. The insulating layer may be formed of a silicon-containing insulator, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, and may be formed by a suitable deposition method such as spin coating, CVD, plasma-enhanced CVD (PECVD), high density plasma CVD (HDP-CVD), or the like. After the substrate **204** is thinned, the exposed surfaces of the conductive vias **208** and the insulating layer (if present) or the substrate **204** are coplanar (within process variations), such that they are level with one another, and are exposed at the back-side **200BS** of the interposer wafer **200**.

(41) After the thinning process, if any, to expose the conductive vias **208**, conductive connectors **134** are formed on the back-side **200BS** of the interposer wafer **200**. The conductive connectors **134** are electrically coupled to the conductive vias **208** and/or integrated circuit dies **50**, **60**, and **70**. The conductive connectors **134** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **134** may be formed using similar materials and methods as the conductive connectors **120** described above with reference to FIG. **4**, and the description is not repeated herein. In the illustrated embodiment, the conductive connectors **134** comprise UBMs **134A**, and solder regions **134B** over the UBMs **134A**. The UBMs **134A** and the solder regions **134B** may be formed using similar material and methods as the UBMs **120A** and the solder regions **120B**, respectively, described above with reference to FIG. **4**, and the description is not repeated herein.

(42) Further, a singulation process is performed by cutting along scribe line regions, e.g., around the package region **200A**. The singulation process may include sawing, etching, dicing, a combination thereof, or the like. For example, the singulation process can include sawing the encapsulant **132**, the interconnect structure **206**, and the substrate **204**. The singulation process singulates the package region **200A** from adjacent package regions to form a singulated package component **400** as illustrated in FIG. **7**. The singulated package component **400** is from the package region **200A**. The singulation process forms interposers **202** from the singulated portions of the interposer wafer **200**. As a result of the singulation process, the outer sidewalls of the interposer **202** and the encapsulant **132** are laterally coterminous (within process variations) as illustrated in FIG. **7**.

(43) FIGS. **8-15** illustrate plan and cross-sectional views of intermediate stages in the manufacturing of a package **600**, in accordance with some embodiments. In particular, FIGS. **8**, **9**, **10A**, **11**, **12**, and **13** illustrate cross-sectional views, FIG. **10B** illustrates a plan view, FIGS. **14** and **15** illustrate magnified views of a region **564** of FIG. **13**.

(44) In FIG. **8**, a package component **400** is placed on a package substrate **500**. The package

substrate **500** includes a substrate core **502**, which may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations thereof, or the like, may also be used. Additionally, the substrate core **502** may be a SOI substrate. Generally, an SOI substrate includes a layer of a semiconductor material such as epitaxial silicon, germanium, silicon germanium, SOI, SGOI, or combinations thereof. In another embodiment, the substrate core **502** is an insulating core such as a fiberglass reinforced resin core. One example core material is fiberglass resin such as FR4. Alternatives for the core material include bismaleimide-triazine (BT) resin, or alternatively, other printed circuit board (PCB) materials or films.

(45) In some embodiments, the substrate core **502** may include active and passive devices (not separately illustrated). Devices such as transistors, capacitors, resistors, combinations thereof, and the like may be used to generate the structural and functional requirements of the design for the system. The devices may be formed using any suitable methods. In some embodiments, the substrate core **502** is substantially free of active and passive devices. In some embodiments, the substrate core **502** further includes conductive vias **504**, which may be also referred to as TSVs. In some embodiments, the conductive vias **504** may be formed using similar materials and methods as the conductive vias **208** described above with reference to FIG. 4, and the description is not repeated herein.

(46) The package substrate **500** may also include a redistribution structure. In some embodiments, the redistribution structure may be formed of alternating layers of dielectric material (e.g., low-k dielectric material) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material, and may be formed through any suitable process (such as deposition, damascene, or the like). In other embodiments, the redistribution structure may be formed of alternating layers of dielectric material (e.g., build up films such as Ajinomoto build-up film (ABF) or other laminates) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material, and may be formed through any suitable process (such as lamination, plating, or the like).

(47) In the illustrated embodiment, the package substrate **500** comprises redistribution structures **506** and **508** formed on opposing surfaces of the substrate core **502**, such that the substrate core **502** is interposed between the redistribution structure **506** and the redistribution structure **508**. The conductive vias **504** electrically couple the redistribution structure **506** to the redistribution structure **508**. In some embodiments, the redistribution structure **506** or the redistribution structure **508** may be omitted.

(48) In some embodiments, bond pads **510** and a solder resist layer **512** are formed on the redistribution structure **506**, with the bond pads **510** being exposed by openings formed in the solder resist layer **512**. The bond pads **510** may be a part of the redistribution structure **506** and may be formed together with other conductive features of the redistribution structure **506**. The solder resist layer **512** may comprise a suitable insulating material (such as a dielectric material, a polymer material, or the like) and may be formed using any suitable deposition methods.

(49) In some embodiments, conductive connectors **514** extend through the opening in the solder resist layer **512** and contact the bond pads **510**. The conductive connectors **514** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **514** may be formed using similar materials and methods as the conductive connectors **120** described above with reference to FIG. 4, and the description is not repeated herein. In the illustrated embodiment, the conductive connectors **514** comprise solder balls.

(50) In some embodiments, bond pads **516** and a solder resist layer **518** are formed on the redistribution structure **508**, with the bond pads **516** being exposed by openings formed in the

solder resist layer **518**. The bond pads **516** may be a part of the redistribution structure **508** and may be formed together with other conductive features of the redistribution structure **508**. The solder resist layer **518** may comprise a suitable insulating material (such as a dielectric material, a polymer material, or the like) and may be formed using any suitable deposition methods.

(51) In some embodiments, conductive connectors **520** extend through the openings in the solder resist layer **518** and contact the bond pads **516**. The conductive connectors **520** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **520** may be formed using similar materials and methods as the conductive connectors **120** described above with reference to FIG. **4**, and the description is not repeated herein. In the illustrated embodiment, the conductive connectors **520** comprise solder balls.

(52) In some embodiments, the package component **400** may be placed on the package substrate **500** using, e.g., a pick-and-place tool. After placing the package component **400** on the package substrate **500**, the conductive connectors **134** are in physical contact with respective conductive connectors **514**, such that the solder regions **134B** of the conductive connectors **134** are in physical contact with the respective conductive connectors **514**.

(53) In FIG. **9**, after placing the package component **400** on the package substrate **500**, a reflow process is performed to mechanically and electrically attach the package component **400** to the package substrate **500**. The reflow process melts and merges the solder regions **134B** of the conductive connectors **134** (see FIG. **8**) and respective solder materials of the conductive connectors **514** (see FIG. **8**) into solder joints **524**. The solder joints **524** electrically and mechanically couple the package component **400** to the package substrate **500**.

(54) In some embodiments, an underfill **526** may be formed around the solder joints **524**, and in a gap between the package component **400** and the package substrate **500**. The underfill **526** may be formed using similar materials and methods as the underfill **126** described above with reference to FIG. **5**, and the description is not repeated herein. In some embodiments, the underfill **526** extends along and physically contacts a sidewall **400L** of the package component **400**, with the sidewall **400L** being opposite to the sidewall **400R**.

(55) FIGS. **10A** and **10B** illustrate a cross-sectional and a plan view after a warpage control structure **530** is attached to the package substrate **500**. The warpage control structure **530** may be attached to the package substrate **500** by an adhesive **528**, such that the adhesive **528** is interposed between the warpage control structure **530** and the solder resist layer **512**. The adhesive **528** may be any suitable adhesive, epoxy, or the like. The warpage control structure **530** may be an annular structure (see FIG. **10B**) and may comprise a hole **531**. The package component **400** may be disposed in the hole **531** of the warpage control structure **530**. The warpage control structure **530** may comprise a metal, a metal alloy, a dielectric material, a semiconductor material, or the like.

(56) In FIG. **11**, a heat dissipation lid **534** is attached to the warpage control structure **530**. In some embodiments, the heat dissipation lid **534** comprises a high thermal conductivity material, such as a metal, a metal alloy, or the like. The heat dissipation lid **534** may be attached to the lid by an adhesive, a thermal interface material, or other means of attaching the two structures. In some embodiments, the heat dissipation lid **534** and the warpage control structure **530** are an integral structure and are placed on the package substrate **500** and the package component **400** at the same time. In some embodiment, a thermal interface material **532** is interposed between the top surface of the package component **400** and the heat dissipation lid **534**. The thermal interface material **532** may comprise a thermal interface material having a high thermal conductivity. In some embodiments, the heat dissipation lid **534** comprises an opening **536** that exposed the lens **74** of the optical engine **70**. This allows for a subsequently formed optical path to have access to the optical engine **70**.

(57) FIG. **12** illustrates a support structure **540** being attached to the solder resist layer **512** of the

package substrate **500** in the opening **536** of the heat dissipation lid **534** using an adhesive **541**. The support structure **540** may comprise a semiconductor material (such as, for example, silicon), a dielectric material, a combination thereof, or the like. The adhesive **541** may be formed using similar materials and methods as the adhesive **528**.

(58) In FIGS. **13**, **14**, and **15** illustrate further processing with FIGS. **14** and **15** illustrating magnified views of a region **564** of FIG. **13**. A lens adapter **550** is attached to the package component **400** and the support structure **540**. The lens adapter **550** provides an interface between the lens **74** of the integrated circuit die **70** and an optical fiber **558** and a mirror **562** that are attached to the lens adapter **550**. The lens adapter **550** may be attached to a top surface of the package component **400** using optical glue **542**, such that the optical glue **542** is in physical contact with the top surface of the integrated circuit die **70** and the top surface of the encapsulant **132**. The lens adapter **550** may be attached to a top surface of the support structure **540** using an optical glue **544**. In some embodiments, the optical glue **542** and **544** comprises a polymer material such as epoxy-acrylate oligomers. The polymer material may have a refractive index between about 1 and about 3. In some embodiments, the optical glue **544** may be replaced with an adhesive similar to adhesive **528** described above.

(59) The lens adapter **550** includes a lens **552** which is over and aligned with the lens **74** of the integrated circuit die **70**. In some embodiments, the lens adapter **550** comprises glass, silicon, the like, or a combination thereof. The mirror **562** is over and attached to the lens adapter **550** with an optical glue **548** and the optical fiber **558** is attached to the lens adapter and the mirror **562** with the optical glue **548**. In some embodiments, the mirror **562** comprises glass, silicon, quartz, acrylic, plastic, the like, or a combination thereof. The mirror **562** allows for a more compact package as it enables a laterally extending optical fiber **558** to utilize the top surface entry of the optical engine **70**. Further, the lens adapter **550** and the support structure **540** enable this compact structure by supporting the optical fiber and the mirror **562**. The optical fiber **558** extends from a mechanical transfer structure **554**. The mechanical transfer structure **554** is attached to the heat dissipation lid **534** by an optical glue **546**. In some embodiments, the optical glue **546** may be replaced with an adhesive similar to adhesive **528** described above.

(60) An optical data path **566** is formed from the optical fiber **558** to the optical coupler **72** of the integrated circuit die **70** by way of the optical glue **548**, the mirror **562**, the lens adapter **550**, the lens **552**, the optical glue **542**, and the lens **74**. The lens **552** of the lens adapter and the lens **74** of the optical engine **70** may each have a dimension in a range from 50 μm to 500 μm with a curvature in a range from 10 μm to 300 μm . The lens **552** may be formed of a dielectric or a semiconductor material, such as silicon, glass, quartz, the like, or a combination thereof.

(61) By having the lens adapter **550**, the support structure **540**, and the mirror **562**, the package structure can be more compact and still utilize a lateral entry optical fiber **558** and top surface entry to the optical engine **70** for the optical data path **566** from the optical fiber **558**.

(62) As shown in FIG. **15**, the mirror **562** has a height **H1**, a width **W1**, and a length (not shown) extending into and out of the cross-sectional view of FIG. **15**. In some embodiments, the length is in a range from 2 mm to 20 mm, the height **H1** is in a range from 0.3 mm to 5 mm, and the width **W1** is in a range from 0.3 mm to 5 mm. The mirror **562** has a reflective surface **562A** that forms a first angle with the incoming optical data path **566** from the optical fiber **558**. In some embodiments, the first angle is substantially 45 degrees (within process variations).

(63) Further as shown in FIG. **15**, the lens adapter has a height **H2**, a width **W2**, and a length (not shown) extending into and out of the cross-sectional view of FIG. **15**. In some embodiments, the length is in a range from 2 mm to 30 mm, the height **H2** is in a range from 0.3 mm to 2 mm, and the width **W2** is in a range from 0.5 mm to 50 mm.

(64) FIGS. **16A-B** illustrate cross-sectional views of the lens adapter **550** and the optical fiber **558** in accordance with some embodiments. The cross-sectional views in FIGS. **16A-B** are along the direction of the length such that the cross-sectional view in FIG. **15** is taken along the A-A lines in

FIGS. 16A-B. FIGS. 16A-B illustrate embodiments with the lens adapter 550 having a grooved top surface with the optical fiber 558 being in one of the grooves. In some embodiments, there are multiple optical fibers 558 on a single lens adapter with each optical fiber 558 being in a separate groove of the lens adapter 550. In these embodiments, the optical fibers 558 can be connected to one or more optical engines 70 of the package structure.

(65) In FIG. 16A, the grooves are V-shaped with the grooves have a depth D1 and the faces of the grooves forming an angle A1 with a vertical line. In some embodiments, the depth D1 is in a range from 50 μ m to 250 μ m, and the angle A1 is substantially 45 degrees (within process variations). In FIG. 16B, the grooves are U-shaped with the grooves having the depth D1.

(66) FIGS. 17A-H illustrated magnified cross-sectional views of the mirror 562 and surrounding components in accordance with some embodiments. Although the prior embodiments have illustrated the mirror 562 as not including any lenses, in some embodiment the mirror 562 can include one or more lens. In FIG. 17A, the mirror 562 includes a convex lens 562B facing the optical fiber 558. In FIG. 17B, the mirror 562 includes a convex lens 562B facing the optical engine 70. In FIG. 17C, the reflective surface 562A of the mirror 562 includes a convex lens 562B. In FIG. 17D, the mirror 562 includes two lenses: a convex lens 562B1 facing the optical fiber 558 and the reflective surface 562A of the mirror 562 includes a convex lens 562B2. In FIG. 17E, the mirror 562 includes a concave lens 562B facing the optical fiber 558. In FIG. 17F, the mirror 562 includes a concave lens 562B facing the optical engine 70. In FIG. 17G, the reflective surface 562A of the mirror 562 includes a concave lens 562B. In FIG. 17H, the mirror 562 includes two lenses: a concave lens 562B1 facing the optical fiber 558 and the reflective surface 562A of the mirror 562 includes a concave lens 562B2.

(67) In each of the embodiments of FIGS. 17A-H, the lens 562B have a dimension in a range from 50 μ m to 500 μ m with a curvature in a range from 10 μ m to 300 μ m. Other embodiments may include alterations of the embodiments of FIGS. 17A-H including multiple concave lenses, combinations of convex and concave lenses, and various other combinations of the embodiments.

(68) The embodiments of the lenses in FIGS. 17A-H show that the design of the lens is flexible, and a combination concave or convex lens design may be utilized based on design requirements. In an embodiment, a combination of any of the lens illustrated in FIGS. 17A-H may be applied based on different alignment requirements. Concave and convex mirror lens have different light guiding properties, and these types of mirror lens can be arranged at any position based on alignment requirements of the design.

(69) FIGS. 18A-C illustrated magnified cross-sectional views of the mirror 562, the optical fiber 558, and surrounding components in accordance with some embodiments. FIG. 18C is a cross-sectional view of the mirror 562 along the length of the mirror 562. In the embodiment of FIGS. 18A-C, the mirror 562 has a recess 562C such that the optical fiber 558 is inserted into the mirror 562. In this embodiment, the optical glue between the mirror 562 and the optical fiber 558 can be omitted as the insertion can hold the components together.

(70) As shown in FIGS. 18B and 18C, the recess 562C of the mirror 562 has a height H3, a length L1, and a depth D2. In some embodiments, the length L1 is in a range from 2 mm to 20 mm, the height H3 is in a range from 0.08 mm to 2 mm, and the depth D1 is in a range from 0.08 mm to 2 mm.

(71) The embodiments of FIGS. 16A-18C can be utilized in any of the package structures in the disclosure.

(72) FIG. 19 illustrates a magnified view of a cross-sectional view of a package component 700, in accordance with some embodiments. The package component 700 is similar to the package component 600 (see FIGS. 13 and 14), with similar features being labeled by similar numerical references, and descriptions of the similar features are not repeated herein. In some embodiments, the package component 700 may be formed using process steps that are similar to the process steps described above with reference to FIGS. 1-14, and the description is not repeated herein. In the

illustrated embodiment, the lens adapter **550** extends from the optical engine **70** to the heat dissipation lid **534**/warpage control structure **530**.

(73) FIG. **20** illustrates a magnified view of a cross-sectional view of a package component **700**, in accordance with some embodiments. The package component **700'** is similar to the package component **700** (see FIG. **19**), with similar features being labeled by similar numerical references, and descriptions of the similar features are not repeated herein. In the illustrated embodiment, the lens adapter **550** extends from the optical engine **70** to the heat dissipation lid **534**/warpage control structure **530** and the support structure **540** is omitted.

(74) FIGS. **21** and **22** illustrate cross-sectional views of package components **800**, in accordance with some embodiments. The package component **800** is similar to the package component **600** (see FIGS. **13** and **14**), with similar features being labeled by similar numerical references, and descriptions of the similar features are not repeated herein. In some embodiments, the package component **800** may be formed using process steps that are similar to the process steps described above with reference to FIGS. **1-14**, and the description is not repeated herein.

(75) In FIG. **21**, a passive component **570** is bonded to the package substrate **500** between the support structure **540** and the heat dissipation lid **534**/warpage control structure **530**. In FIG. **22**, the passive component **570** is bonded to the package substrate **500** between the support structure **540** and the package component **400**. The passive component **570** is below the optical fiber **558**.

(76) The passive component **570** may include one or more passive devices such as capacitors, resistors, inductors, or the like. The passive component **570** may be formed in a similar manner to the integrated circuit die **50**, and the description is not repeated herein. The passive component **570** may be bonded to conductive features of the package substrate **500** by connectors **572**. The connectors **572** may be similar to the conductive connectors **134** described above and the description is not repeated herein. The passive component **570** may be bonded to the package substrate **500** and have an underfill **574** surrounding the connectors **572**. The bonding process of the passive component **570** may be similar to the bonding process of the package component **400** described above and the description is not repeated herein. The underfill **574** may be similar to the underfill **526** described above and the description is not repeated herein.

(77) FIG. **23** illustrates a cross-sectional view of a package component **900**, in accordance with some embodiments. The package component **900** is similar to the package component **600** (see FIGS. **13** and **14**), with similar features being labeled by similar numerical references, and descriptions of the similar features are not repeated herein. In some embodiments, the package component **900** may be formed using process steps that are similar to the process steps described above with reference to FIGS. **1-14**, and the description is not repeated herein. In the illustrated embodiment, the package component **400** is a multi-chip module with the optical engine **70** be a separate module from the dies **50** and **60**. In this embodiment, the package including the dies **50** and **60** and the optical engine **70** are bonded separately to the package substrate **500**.

(78) FIG. **24** illustrates a cross-sectional view of a package component **1000**, in accordance with some embodiments. The package component **1000** is similar to the package component **600** (see FIGS. **13** and **14**), with similar features being labeled by similar numerical references, and descriptions of the similar features are not repeated herein. In some embodiments, the package component **1000** may be formed using process steps that are similar to the process steps described above with reference to FIGS. **1-14**, and the description is not repeated herein. In the illustrated embodiment, the package component **400'** is an integrated fan-out (InFO) package instead of a chip-on-wafer package. For example, in this embodiment, the interposer **202** is replaced with a redistribution structure **200'**.

(79) In some embodiments, the redistribution structure **200'** may be formed of alternating layers of dielectric material (e.g., low-k dielectric material) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material, and may be formed through any suitable process (such as deposition, damascene, or the like). In other embodiments, the redistribution structure **200'**

may be formed of alternating layers of dielectric material (e.g., build up films such as Ajinomoto build-up film (ABF) or other laminates) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material, and may be formed through any suitable process (such as lamination, plating, or the like).

(80) FIG. 25 illustrates a magnified view of a cross-sectional view of a package component **1100**, in accordance with some embodiments. The package component **1100** is similar to the package component **600** (see FIGS. 13 and 14), with similar features being labeled by similar numerical references, and descriptions of the similar features are not repeated herein. In some embodiments, the package component **1100** may be formed using process steps that are similar to the process steps described above with reference to FIGS. 1-14, and the description is not repeated herein. In the illustrated embodiment, the lens **74** on the optical engine **70** is recessed into the optical engine **70**.

(81) Embodiments may achieve advantages. By forming a package (such as, for example, the package component **600** illustrated in FIGS. 13 and 14) comprising a lens adapter, a support structure, and a mirror as described above various advantages may be achieved. By including these components in specific configurations, the package structure can be more compact and utilize a lateral entry optical fiber to the package structure with a top surface entry to the optical engine for the optical data path from the optical fiber. Accordingly, improved coupling between the coupler of the optical integrated circuit die and an optical fiber is achieved. Various embodiments presented herein allow for integration of optical integrated circuit dies comprising grating couplers, achieving high bandwidth with ultra-low power consumption through the edge coupler, and extensive integration for co-package optics. Further, the optical loss from the optical fiber to the mirror is reduced as compared to other structures. In addition, the lens adapter and the mirror allow for the disclosed embodiments to be widely usable in various packaging configurations, such as multi-chip modules (MCM), chip-on-wafer-on-substrate packages, or integrated fan-out (InFO) packages.

(82) An embodiment is a package including a package substrate. The package also includes a package component bonded to the package substrate, the package component including an interposer. The package also includes an optical die bonded to the interposer, the optical die including an optical coupler. The package also includes an integrated circuit die bonded to the interposer adjacent the optical die. The package also includes a lens adapter adhered to the optical die with a first optical glue. The package also includes a mirror adhered to the lens adapter with a second optical glue, the mirror being aligned with the optical coupler of the optical die. The package also includes an optical fiber on the lens adapter, a first end of the optical fiber facing the mirror, the optical fiber being configured such that an optical data path extends from the first end of the optical fiber through the mirror, the second optical glue, the lens adapter, and the first optical glue to the optical coupler of the optical die.

(83) Embodiments may include one or more of the following features. The package where the optical die includes a first lens on a backside of the optical die, the backside of the optical die facing away from the package substrate, the optical data path going through the first lens. The lens adapter includes a second lens on a lower surface of the lens adapter, the optical data path going through the second lens. The mirror includes a third lens, the optical data path going through the third lens. The mirror is made of glass, silicon, quartz, acrylic, plastic, or a combination thereof. The package further including a support structure adhered to the package substrate, the lens adapter being over and adhered to the support structure. The heat dissipation lid includes an opening, a portion of the optical die being in the opening, the lens adapter being adhered to the portion of the optical die in the opening. The lens adapter extends across the opening in the heat dissipation lid and is adhered to an outer portion of the heat dissipation lid. The package further including a second underfill between the package substrate and the package component. The package component further includes a first underfill between the optical die, the integrated circuit die, and the interposer, and an encapsulant encapsulating the optical die, the integrated circuit die, and the

first underfill, the lens adapter being adhered to the encapsulant. The lens adapter has grooves in a top surface of the lens adapter, the optical fiber being in one of grooves. The package where the mirror includes a recess, the optical fiber being in the recess of the mirror.

(84) An embodiment is a package including a package substrate. The package also includes a package component bonded to the package substrate. The package also includes a heat dissipation lid attached to the package substrate and covering the package component, the heat dissipation lid including an opening, a portion of the package component being in the opening of the heat dissipation lid. The package also includes a lens adapter in the opening of the heat dissipation lid, the lens adapter being attached to the package component. The package also includes a mirror in the opening of the heat dissipation lid, the mirror being attached to the lens adapter. The package also includes an optical fiber extending to the opening in the heat dissipation lid, the optical fiber being on the lens adapter, where the package component includes an optical die including an optical coupler, the optical coupler being proximate a first sidewall of the optical die, the first sidewall of the optical die and the first sidewall of the package component in the opening of the heat dissipation lid, a first end of the optical fiber facing the mirror, the optical fiber being configured such that an optical data path extends from the first end of the optical fiber through the mirror and the lens adapter to the optical coupler of the optical die.

(85) Embodiments may include one or more of the following features. The package further including a support structure on the package substrate in the opening of the heat dissipation lid, the lens adapter being attached to the support structure. The optical die includes a first lens on a backside of the optical die, the backside of the optical die facing away from the package substrate, the lens adapter including a second lens on a lower surface of the lens adapter, and the mirror including a third lens, the optical data path going through the first lens, the second lens, and the third lens. The optical data path is reflected in the mirror. The package component further includes an interposer, the optical die being bonded to the interposer, a logic die bonded to the interposer, a memory die bonded to the interposer, a first underfill between the optical die, the logic die, the memory die, and the interposer, and an encapsulant encapsulating the optical die, the logic die, the memory die, and the first underfill, the lens adapter being attached to the encapsulant.

(86) An embodiment includes a method of forming a package component, where forming the package component includes bonding an optical die to an interposer, the optical die including an optical coupler proximate to a first sidewall of the optical die. The method also includes bonding a logic die to the interposer. The method also includes depositing a first underfill between the optical die, the logic die, and the interposer. The method also includes forming an encapsulant on the optical die, the logic die, and the first underfill. The method also includes bonding the package component to a package substrate. The method also includes attaching a heat dissipation lid to the package substrate and the package component, the heat dissipation lid including an opening, a portion of the package component being in the opening of the heat dissipation lid. The method also includes attaching a support structure to the package substrate in the opening of the heat dissipation lid. The method also includes attaching a lens adapter to the portion of the package component in the opening and the support structure in the opening. The method also includes attaching a mirror to the lens adapter. The method also includes attaching an optical fiber to the mirror, the optical fiber being on the lens adapter, the optical fiber being configured such that an optical data path extends from a first end of the optical fiber through the mirror and the lens adapter to the optical coupler of the optical die.

(87) Embodiments may include one or more of the following features. The method where the optical die includes a first lens on a backside of the optical die, the backside of the optical die facing away from the package substrate, the lens adapter including a second lens on a lower surface of the lens adapter, and the mirror including a third lens, the optical data path going through the first lens, the second lens, and the third lens. The lens adapter extends across the opening in the heat dissipation lid and is attached to an outer portion of the heat dissipation lid.

(88) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A package comprising: a package substrate; a package component bonded to the package substrate, the package component comprising: an interposer; an optical die bonded to the interposer, the optical die comprising an optical coupler; and an integrated circuit die bonded to the interposer adjacent the optical die; a lens adapter adhered to the optical die with a first optical glue; a mirror adhered to the lens adapter with a second optical glue, the mirror being aligned with the optical coupler of the optical die; and an optical fiber on the lens adapter, a first end of the optical fiber facing the mirror, the optical fiber being configured such that an optical data path extends from the first end of the optical fiber through the mirror, the second optical glue, the lens adapter, and the first optical glue to the optical coupler of the optical die.
2. The package of claim 1, wherein the optical die includes a first lens on a backside of the optical die, the backside of the optical die facing away from the package substrate, the optical data path going through the first lens.
3. The package of claim 2, wherein the lens adapter includes a second lens on a lower surface of the lens adapter, the optical data path going through the second lens.
4. The package of claim 2, wherein the mirror includes a third lens, the optical data path going through the third lens.
5. The package of claim 1, wherein the mirror is made of glass, silicon, quartz, acrylic, plastic, or a combination thereof.
6. The package of claim 1 further comprising: a support structure adhered to the package substrate, the lens adapter being over and adhered to the support structure.
7. The package of claim 1 further comprising: a heat dissipation lid attached to the package substrate and covering the package component, wherein the heat dissipation lid comprises an opening, a portion of the optical die being in the opening, the lens adapter being adhered to the portion of the optical die in the opening.
8. The package of claim 7, wherein the lens adapter extends across the opening in the heat dissipation lid and is adhered to an outer portion of the heat dissipation lid.
9. The package of claim 1, wherein the package component further comprises: a first underfill between the optical die, the integrated circuit die, and the interposer; and an encapsulant encapsulating the optical die, the integrated circuit die, and the first underfill, the lens adapter being adhered to the encapsulant.
10. The package of claim 8 further comprising: a second underfill between the package substrate and the package component.
11. The package of claim 1, wherein the lens adapter has grooves in a top surface of the lens adapter, the optical fiber being in one of the grooves.
12. The package of claim 1, wherein the mirror comprises a recess, the optical fiber being in the recess of the mirror.
13. A package comprising: a package substrate; a package component bonded to the package substrate; a heat dissipation lid attached to the package substrate and covering the package component, the heat dissipation lid comprising an opening, a portion of the package component

being in the opening of the heat dissipation lid; a lens adapter in the opening of the heat dissipation lid, the lens adapter being attached to the package component; a mirror in the opening of the heat dissipation lid, the mirror being attached to the lens adapter; and an optical fiber extending to the opening in the heat dissipation lid, the optical fiber being on the lens adapter, wherein the package component comprises: an optical die comprising an optical coupler, the optical coupler being proximate a first sidewall of the optical die, the first sidewall of the optical die and a first sidewall of the package component in the opening of the heat dissipation lid, a first end of the optical fiber facing the mirror, the optical fiber being configured such that an optical data path extends from the first end of the optical fiber through the mirror and the lens adapter to the optical coupler of the optical die.

14. The package of claim 13 further comprising: a support structure on the package substrate in the opening of the heat dissipation lid, the lens adapter being attached to the support structure.

15. The package of claim 14, wherein the optical die includes a first lens on a backside of the optical die, the backside of the optical die facing away from the package substrate, the lens adapter including a second lens on a lower surface of the lens adapter, and the mirror including a third lens, the optical data path going through the first lens, the second lens, and the third lens.

16. The package of claim 13, wherein the optical data path is reflected in the mirror.

17. The package of claim 13, wherein the package component further comprises: an interposer, the optical die being bonded to the interposer; a logic die bonded to the interposer; a memory die bonded to the interposer; a first underfill between the optical die, the logic die, the memory die, and the interposer; and an encapsulant encapsulating the optical die, the logic die, the memory die, and the first underfill, the lens adapter being attached to the encapsulant.

18. A method comprising: forming a package component, wherein forming the package component comprises: bonding an optical die to an interposer, the optical die comprising an optical coupler proximate to a first sidewall of the optical die; bonding a logic die to the interposer; depositing a first underfill between the optical die, the logic die, and the interposer; and forming an encapsulant on the optical die, the logic die, and the first underfill; bonding the package component to a package substrate; and attaching a heat dissipation lid to the package substrate and the package component, the heat dissipation lid comprising an opening, a portion of the package component being in the opening of the heat dissipation lid; attaching a support structure to the package substrate in the opening of the heat dissipation lid; attaching a lens adapter to the portion of the package component in the opening and the support structure in the opening; attaching a mirror to the lens adapter; and attaching an optical fiber to the mirror, the optical fiber being on the lens adapter, the optical fiber being configured such that an optical data path extends from a first end of the optical fiber through the mirror and the lens adapter to the optical coupler of the optical die.

19. The method of claim 18, wherein the optical die includes a first lens on a backside of the optical die, the backside of the optical die facing away from the package substrate, the lens adapter including a second lens on a lower surface of the lens adapter, and the mirror including a third lens, the optical data path going through the first lens, the second lens, and the third lens.

20. The method of claim 18, wherein the lens adapter extends across the opening in the heat dissipation lid and is attached to an outer portion of the heat dissipation lid.
